SHORLYNOT

Quantum-Inspired Cyber Threat Detection for Digital Signature Security

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Executive Summary: ShorlyNot is an end-to-end quantum security architecture solving classical digital signature vulnerability against quantum forgery and harvesting attacks. It replaces classical algorithms (RSA/ECDSA/Ed25519) with **teleportation-based Quantum Digital Signatures (ShorlyNot-QDS-T1)** running on real Qiskit Aer quantum circuits, protected by BB84 QKD and PQC layers, with real-time **non-machine-learning threat detection (Q-STDF)** governed by Hoeffding statistical bounds.

1. System Architecture: Exactly Two Products

• **skeleton/ (The Core Invention):** Standalone quantum framework & FastAPI service (:8000). Executes 3-qubit Bell teleportation circuits on Qiskit Aer, statistical bounding, and BB84 QKD.
• **bank/ (The Operational Proof):** Full banking web app (:8080) & real-time dark SOC console. Every transaction is quantum-signed; detected threats automatically trigger application lockdown (S0 to S4).

2. ShorlyNot-QDS-T1 Teleportation Protocol Specification

Stage	Signer (Alice) / Verifier (Bob) Action	Underlying Quantum Physics / Math
1. Encode	Hash payload (SHA-256) & derive L=128 check positions.	Pauli states |0>, |1>, |+>, |-> via {Z, X} bases
2. Entangle	Distribute Bell pairs |Φ+> across quantum channel.	|Φ+> = (|00> + |11>) / √2
3. Teleport	Alice executes Bell-state measurement (BSM) circuit.	Yields 2 classical syndrome bits (m1, m2) ∈ {0,1}²
4. Protect	Encrypt syndromes with BB84 QKD session key.	PQC AES-256-GCM authenticated payload wrapper
5. Correct	Bob applies Pauli correction matrix U(m1, m2) to his qubit.	U ∈ {I, X, Z, XZ} restores original state
6. Verify	Bob projects in Alice's basis β & checks error rate p■.	Accepted if and only if p■ ≤ τ (Hoeffding Bound)

3. Pauli Teleportation Unitary Lookup Matrix

Syndrome (m1, m2)	Pauli Operator U	Matrix Representation	Physical Action on Bob's Entangled State
(0, 0)	I (Identity)	[[1, 0], [0, 1]]	No transformation required
(0, 1)	X (Bit Flip)	[[0, 1], [1, 0]]	Flips |0> ↔ |1>
(1, 0)	Z (Phase Flip)	[[1, 0], [0, -1]]	Inverts relative phase |+> ↔ |->
(1, 1)	XZ (Bit + Phase)	[[0, -1], [1, 0]]	Simultaneous bit & phase correction

SHORLYNOT · Threat Engine & Mathematical Proofs

4. Mathematical Formulation: Hoeffding Statistical Bound

The deterministic acceptance threshold τ is computed from Hoeffding's Inequality to guarantee provable bounds against forgery while allowing honest physical channel noise:

$$\tau = p0 + \sqrt{\frac{\ln(1/\delta)}{2n}}$$

- **Honest Noise Floor (p0):** 0.02 (2.0%) · **Check Qubits (n):** 64 · **False Rejection Budget (δ):** 0.01 (1.0%).
- **Calculated Threshold:** $\tau = 0.02 + \sqrt{(\ln(100)/128)} =$ **0.2097** (~21.0%).
- **Honest Signer:** p■ ≈ 0.00 to 0.02 ≤ τ (Accepted).
- **Random Forger:** p■ ≈ 0.50 > τ (Probability of bypassing threshold is P_forge < 9.40 × 10&supmin;&sup7;).

5. Q-STDF 6-Step Non-ML Threat Decision Ladder

Attack Vector	Decision Step & Deterministic Check	Mathematical / Protocol Guarantee	Stage Escalation
UNAUTH_VERIFY	Step 1: Check verifier identity credentials	Role-based cryptographic entitlement check	Stage S3 (Reset)
IMPERSONATION	Step 2: Signer identity != registered key ID	Hardware-bound identity certificate validation	Stage S2 (Block)
REPLAY	Step 3: Nonce exists in replay LRU cache	Cryptographic uniqueness within time window	Stage S3 (Reset)
CHANNEL	Step 4: PQC tag validation or QBER > 11%	Eavesdropping breaks entanglement / flips bits	Stage S4 (Lock)
FORGERY	Step 5: Mismatch rate p■ > τ (Hoeffding)	Quantum state disturbance cannot be forged	Stage S2 → S4
OK	Step 6: All quantum & protocol checks passed	Empirical error p■ ≤ 0.2097 (Honest proof)	Stage S0 (Normal)

6. Application Security Stages (S0 to S4)

Stage	State Name	Triggering Condition	Bank Application Enforcement Action
S0	Normal	All checks p■ ≤ τ, QBER < 5%	Full operations active; transactions committed ■
S1	Anomaly Watch	QBER 5%-8% or single soft noise	Soft operator alert; transactions permitted
S2	Threat Active	Signature Forgery or Impersonation	New transfers blocked at application level ■
S3	Pattern Alert	Replay attack or unauthorized verifier	Read-only ledger view; user sessions purged
S4	Critical Lockdown	Channel tampering or repeated forgeries	Full application lockout page; requires ops reset ■

7. North Star 90-Second Live Demo Script

1. Launch scripts/run_demo.bat → opens Bank (:8080) and Skeleton (:8000).
2. **Honest Transfer:** Login as Alice (alice/alice123), send ■5,000 to Bob → Qiskit Aer circuit runs, p■ = 0.00 ≤ 0.2097, balance updates.
3. **Attack & Threshold Catch:** Open SOC at /soc, trigger 'FORGERY' → mismatch p■ ≈ 0.50 > 0.2097, Stage escalates to S2.
4. **Application Block:** Return to Bank, attempt transfer → Refused by Security Policy with parameters displayed.
5. **Full Lockdown & Reset:** Trigger 'CHANNEL' on SOC → Stage escalates to S4, Bank displays lockout screen → click Reset to restore S0.